

```
import cv2
import matplotlib.pyplot as plt

# Load noisy image
image = cv2.imread("noisy_image.jpg")

if image is None:
    print("Image not found!")
    exit()

# Convert BGR to RGB
image = cv2.cvtColor(image, cv2.COLOR_BGR2RGB)

# Noise Reduction using Gaussian Blur
gaussian = cv2.GaussianBlur(image, (5, 5), 0)

# Noise Reduction using Median Filter
median = cv2.medianBlur(image, 5)

# Display Images
plt.figure(figsize=(12,4))

plt.subplot(1,3,1)
plt.imshow(image)
plt.title("Original Noisy Image")
plt.axis("off")

plt.subplot(1,3,2)
plt.imshow(gaussian)
plt.title("Gaussian Filter")
plt.axis("off")

plt.subplot(1,3,3)
plt.imshow(median)
plt.title("Median Filter")
plt.axis("off")
```

```
plt.tight_layout()
plt.show()
```

Image not found!

```
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error                                Traceback (most recent call last)
/tmp/ipykernel_23726/1614071750.py in <cell line: 0>()
    10
    11 # Convert BGR to RGB
--> 12 image = cv2.cvtColor(image, cv2.COLOR_BGR2RGB)
    13
    14 # Noise Reduction using Gaussian Blur

error: OpenCV(4.13.0) /io/opencv/modules/imgproc/src/color.cpp:199: error: (-215:Assertion failed)
!_src.empty() in function 'cvtColor'
```

Next steps: [Explain error](#)

